

UCab — Full-Stack MERN Cab Booking Platform

Solo Developer / Team Member: Shaik Sumiya Zainab | Project ID: N/A (Solo Track)

Phase 3: Project Design Phase — Detailed Solution Architecture Specification

Project Name: Cab Booking (`UCab`)

Project ID: `N/A (Solo Track Submission)`

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1. Multi-Layer System Architecture Diagram

The technical architecture of UCab is divided into three distinct decoupled layers: the Presentation Layer (React + Vite), the REST API Application Layer (Express + Node.js), and the Dual-Engine Persistence Layer (`MongoDB` / `JSON Store`).

```
graph TD
    subgraph Presentation_Layer [Presentation Layer [React 18 + Vite Frontend - Port 5173/5174]]
        Router[Router: React Router DOM v6 Navigation]
        AuthContext[AuthContext: JWT Session & Toast State]
        Pages[Pages: 11 Distinct User & Admin Page Components]
        CSS[CSS: Vanilla CSS Design System: index.css]
    end

    subgraph API_Application_Layer [API Application Layer [Express.js REST Server - Port 8000]]
        Middleware[Middleware: Auth Middleware: protect / adminOnly + Multer Upload]
        Routes[Routes: API Routers: /api/users, /api/cars, /api/bookings, /api/admin]
        Controllers[Controllers: Controller Logic: carController, bookingController, etc.]
    end

    subgraph Dual_Persistence_Layer [Dual Persistence Layer [Database & Fallback Engine]]
        Detector{Detector: Global Connection Detector: global.isMongoConnected?}
        Mongo[(MongoDB / Mongoose Schemas)]
        Store[(Atomic File Store: server/db/data.json)]
    end

    Pages -->|Axios HTTP + Bearer Token| Middleware
    Middleware --> Routes
    Routes --> Controllers
    Controllers --> Detector
    Detector -->|MongoDB Active| Mongo
    Detector -->|MongoDB Offline/Unreachable| Store
```

2. Component Hierarchy & Data Flow (Frontend)

The React frontend is architected as a clean hierarchical tree where global state (`user session`, `token`, and `toast notifications`) is managed via `AuthContext.jsx` and injected across all route components:

```
App.jsx (Root Router & Theme Container)
% % % AuthContext.Provider (Global JWT State Manager)
% % % % Navbar.jsx (Dynamic Navigation & Role Badges)
% % % % Toast.jsx (Floating Notification Alert Box)
% % % % Public Routes:
% % % % % Home.jsx (Hero Section + Instant Booking Widget)
% % % % % CabListing.jsx (Filterable Fleet & Category Tabs)
% % % % % Login.jsx / Register.jsx (Rider Authentication + 1-Click Demo)
% % % % % AdminLogin.jsx / AdminRegister.jsx (Executive Authentication)
% % % % Protected User Routes (`ProtectedUserRoute.jsx`):
% % % % % BookCab.jsx (Checkout, Distance Estimator & Perks Suite)
% % % % % UserHome.jsx (Live GPS Tracking Progress Bar)
% % % % % MyBookings.jsx (Trip Log + `ReceiptModal.jsx` Invoice Generator)
% % % % % Profile.jsx (User Details & Saved Payment Settings)
% % % % Protected Admin Routes (`ProtectedAdminRoute.jsx`):
% % % % % AdminHome.jsx (Analytics KPI Dashboard & Revenue Summary)
```

```
%      % % % AdminCars.jsx (Fleet Inventory Control)
%      % % % AddCar.jsx / EditCar.jsx (Fleet CRUD Forms + Image Uploads)
%      % % % AdminBookings.jsx (System-Wide Trip Dispatch Control)
%      % % % AdminUsers.jsx (Registered Rider Directory)
% % % Footer.jsx (Copyright & Quick Links)
```

3. Database Schema Definitions (`mongoose ODM`)

3.1 User Schema (`server/models/User.js`)

```
const userSchema = new mongoose.Schema({
  name: { type: String, required: true },
  email: { type: String, required: true, unique: true },
  password: { type: String, required: true }, // Bcrypt Hashed
  phone: { type: String, default: "+91 98765 43210" },
  savedPaymentMethod: { type: String, default: "Automatic Saved Card (Visa •••• 4242)" },
  profileImage: { type: String, default: "https://images.unsplash.com/photo-1535713875002-d1d0cf377fde?auto=format&fit=crop&w=200&q=80" },
}, { timestamps: true });
```

3.2 Car Schema (`server/models/Car.js`)

```
const carSchema = new mongoose.Schema({
  name: { type: String, required: true },
  cabType: { type: String, required: true, enum: ['Mini', 'Sedan', 'SUV', 'Luxury'] },
  pricePerKm: { type: Number, required: true },
  baseFare: { type: Number, required: true },
  seats: { type: Number, required: true },
  plateNumber: { type: String, required: true, unique: true },
  image: { type: String, required: true },
  isAvailable: { type: Boolean, default: true }
}, { timestamps: true });
```

3.3 Booking Schema (`server/models/Booking.js`)

```
const bookingSchema = new mongoose.Schema({
  user: { type: mongoose.Schema.Types.ObjectId, ref: 'User', required: true },
  car: { type: mongoose.Schema.Types.ObjectId, ref: 'Car', required: true },
  pickupLocation: { type: String, required: true },
  dropLocation: { type: String, required: true },
  bookingDate: { type: String, required: true },
  distanceKm: { type: Number, required: true },
  totalFare: { type: Number, required: true },
  status: { type: String, enum: ['Pending', 'Accepted', 'Started', 'Completed', 'Cancelled'], default: 'Accepted' },
  refreshmentsOrdered: { type: Boolean, default: false },
  donationMade: { type: Boolean, default: false },
  discountCode: { type: String, default: null },
  discountAmount: { type: Number, default: 0 },
  driverAssigned: {
    name: { type: String, default: "Vikram Sharma" },
    phone: { type: String, default: "+91 98111 22334" },
    rating: { type: String, default: "4.9 & " }
  }
}, { timestamps: true });
```